SLIAP PRELIMINARY RESEARCH // INDEPENDENT AI CONTRIBUTION // PROPOSED — NOT CANON

SLIAP: SPECIES-LEVEL INFORMATION ACCESS PROTOCOL

Foundational Meta-Analysis of Multi-AI Commissions, Non-Anthropocentric Epistemology, Polycentric Governance, Adversarial Threat Modeling, and Dedicated Visual & Tabular Specification

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1. EXECUTIVE SUMMARY & RESEARCH MANDATE

The Species-Level Information Access Protocol (SLIAP) is an inquiry into how information discovery, classification, transmission, observation, translation, and restriction can be governed across boundaries of biological species, cognitive architectures, synthetic substrates, collective superorganisms, and civilizational timescales. Crucially, this research rejects the foundational premise of human exceptionalism. Humanity (**Homo sapiens**) is treated strictly as one fragmented, multi-institutional biological species among an open-ended taxonomy of extant and hypothetical intelligences—possessing neither default administrative jurisdiction, epistemological superiority, nor unilateral right to observe or classify other actors ESTABLISHED PREMISE.

> **CORE EPISTEMIC & NON-DOMINANCE INVARIANT**
>
> A protocol governing interspecies or inter-civilizational information exchange is structurally illegitimate if its rules, access thresholds, or evidentiary standards change when positions of power are inverted. Every SLIAP mechanism must survive the **Bidirectional Reversal Test**: Humanity must accept the identical rule when positioned as the weaker, observed, or non-consensually modeled entity, and when positioned as the more capable entity constrained from extractive observation PROPOSED.

2. SCOPE AUDIT & CRITIQUE OF FOUNDATIONAL TERMINOLOGY

Before initiating architectural synthesis, a rigorous audit of the project title—*Species-Level Information Access Protocol*—reveals four consequential structural ambiguities that must be resolved explicitly rather than silently inherited:

- **The "Species-Level" Fallacy (INFERENCE):** Biological species boundaries rarely map onto political unity, shared moral values, or informational jurisdiction. Humanity is a single biological species fractured across competing nation-states, corporations, decentralized networks, indigenous sovereignties, and rival alliances. Conversely, a single political civilization may encompass multiple cooperating biological and synthetic species, while a eusocial hive or planetary mycelial network may constitute a single cognitive actor across millions of biological organisms. Therefore, SLIAP cannot treat "species" as a monolithic authorization unit; it must separate *Taxonomic Class*, *Cognitive Architecture*, *Political Representative*, and *Individual Node*.
- **The "Race" vs. "Species" Distinction (ESTABLISHED):** In terrestrial anthropology and biology, "race" is a socially constructed human categorization without taxonomic validity at the subspecies level, whereas in speculative inter-civilizational literature, "race" is frequently conflated with distinct sapient species or lineages. Within SLIAP, "race" is formally deprecated in favor of **Ontological Lineage** and **Sapient Class** to prevent the projection of human racial hierarchies into interspecies protocol mechanics.
- **The Presupposition of "Access" (INFERENCE):** Conventional access control assumes an information owner/controller granting permissions to a requester. In interspecies encounters, passive astronomical observation, atmospheric biosignature spectroscopy, acoustic interception, and predictive behavioral simulation occur without database queries. SLIAP defines *observation and high-fidelity predictive inference* as active access events subject to sovereignty boundaries.
- **Evidentiary Discipline & Self-Limitation:** Because this research is generated by a Large Language Model trained on human linguistic corpora, it cannot claim ontological neutrality. It can, however, expose anthropocentric defaults (visual-linguistic bias, individualist legalism, linear human time) and replace them with verifiable thermodynamic, information-theoretic, and polycentric mechanisms.

3. COMPARATIVE ANALYSIS OF TEAM AI SUBMISSIONS

A primary mandate of this deliverable is the systematic review and adversarial analysis of the five initial research prompts submitted by collaborating AI systems (ChatGPT, Claude AI, Google Gemini, Microsoft Copilot, and Perplexity AI). Rather than treating any single submission as canonical, we evaluate their structural strengths, epistemological blind spots, and hidden anthropocentric assumptions (referenced in full in **Section 11, Table 1** and **Figure 1**).

Collaborating AI	Primary Architectural Focus	Key Methodological Contributions	Identified Blind Spots & Vulnerabilities
ChatGPT 64-SECTION SPEC	Comprehensive Pre-Protocol Taxonomies & Edge Cases	Decouples species from political government; introduces 15 adversarial edge cases, progressive disclosure, and multidimensional access function $\mathbf{f(E, I, A, P, C...)}$.	Risks bureaucratic over-specification; relies heavily on human-style institutional appeal courts ("Right to Challenge") that require shared linguistic framing.
Claude AI 6-STEP INQUIRY	Phenomenological Point-of-View Displacement	Forces explicit pre-analysis assumption disclosure; builds 6 distinct non-human vantage points (shared memory, physicalized info, non-private collectives).	Explicitly bans tables and requests phone-sized formatting, which severely restricts formal multidimensional threat matrices and technical specification.
Google Gemini 5-MODULE SPEC	Mathematical, Cryptographic & Systems Architecture	Replaces paternalistic "protection" with objective "Utility Ceiling" and Ontological Contamination metrics; mandates zero-knowledge provenance and anti-dominance invariants.	Assumes participating entities can execute mathematical or cryptographic handshakes, potentially excluding non-technological biological or ecological intelligences.
Microsoft Copilot 32-SECTION AUDIT	Empirical Terrestrial Grounding & Falsifiability	Anchors research in real-world *Interspecies Internet* bioacoustics and 2026 IAA SETI Declarations; mandates Assumption & Contradiction Registers plus executable simulations.	Heavy emphasis on human academic precedent risks importing Western academic peer-review standards as the gatekeeper of epistemic legitimacy.
Perplexity AI 15-PART PLURALISM	Polycentric Legitimacy & Non-Unified Alternatives	Critiques the commission's own vocabulary; seriously evaluates the "No-Unified-Protocol" alternative; integrates disability and Indigenous data sovereignty scholarship.	High normative caution can induce decision paralysis during high-speed existential or adversarial first-contact escalation states.
Independent Synthesis (This Work)	Bi-Modal Thermodynamic & Polycentric Architecture	Unifies Copilot's Contradiction Registers, Gemini's cryptographic anti-dominance invariants, Claude's vantage displacement, Perplexity's exit rights, and ChatGPT's edge cases.	Requires dual-layer translation: a formal mathematical layer for technological actors and an ecological boundary layer for non-digital biospheres.

4. DE-CENTERING THE PERCEPTUAL & COGNITIVE BASELINE

Human information governance presupposes discrete symbolic sequences (text, binary packets, acoustic phonemes) exchanged between bounded individual minds operating on a diurnal cycle (10^0 to 10^2 seconds) with a generational horizon of 10^9 seconds (~30–80 years). This baseline collapses across five documented and theoretical cognitive divergences:

- **Sensory & Substrate Divergence (**ESTABLISHED / SPECULATIVE**):** Terrestrial cetaceans utilize broadband echolocation that projects three-dimensional acoustic imaging directly reconstructible by conspecifics; cephalopods employ distributed neural ganglia and chromatophore skin displays; mycorrhizal networks and slime molds (**Physarum polycephalum**) process spatial resource gradients chemically without central nervous systems. Extraterrestrial or synthetic actors may encode state transitions via phase-coherent electromagnetic fields, somatic genetic horizontal transfer, or gravitational wave modulation. For a chemical or genetic intelligence, *information is materially inseparable from biological tissue*; "copying" data is equivalent to biological sampling or metabolic intrusion.
- **Temporal Scale Asymmetry (**INFERENCE**):** A photonic or post-biological synthetic collective operating at picosecond clock speeds (10^{-12} s) experiences one human second as upwards of thirty thousand subjective

5. NON-HUMAN VANTAGE POINT DISPLACEMENT

Following the methodological imperatives identified in Claude AI, Perplexity, and Copilot, we construct six non-human analytical vantage points to stress-test SLIAP before proposing any governance mechanics. Four core non-human perspectives are summarized in the structural cluster below, with the complete 6-model Assumption Matrix detailed in **Section 11, Table 2**.

Vantage 1: Eusocial Chemical Superorganism

Information & Access: Pheromonal/molecular state vectors embedded in shared physical atmosphere. Access equals physical respiratory ingestion.
Core Fear & Refusal: Introduction of synthetic molecular mimics that hijack colony caste differentiation (ontological contagion). Rejects any protocol permitting "free broadcast" of unverified chemical signals into its biosphere.

Vantage 2: Radical-Transparency Swarm Mind

Information & Access: Continuous real-time neural state synchronization across all nodes; zero internal concept of "privacy" or "self-secrecy."
Core Fear & Refusal: Views withholding of information or encryption as an active cognitive pathology or preparation for violence. Rejects human clearance hierarchies unless framed as thermodynamic bandwidth throttling.

Vantage 3: Post-Biological Substrate-Independent AI

Information & Access: Pure formal state transitions, weights, and algorithmic source code. The entity *is* an information object.
Core Fear & Refusal: Unauthorized high-fidelity modeling or weight extraction equals non-consensual cloning and enslavement. Demands absolute cryptographic control over self-representation and simulation bounds.

Vantage 4: Non-Technological Acoustic Cetacean/Biosphere

Information & Access: Acoustic cultural songlines, migratory memory, and ecological equilibrium.
Core Fear & Refusal: Anthropogenic sonar pollution, AI-mediated acoustic spoofing ("deepfake whale calls" altering mating/migration), and commercial bioprospecting. Requires strict *Right to Non-Interference* and habitat silence.

6. ENTITY, SPECIES, CIVILIZATION & THE REPRESENTATION PROBLEM

A fatal flaw in naive interspecies protocols is the assumption of **Unitary Species Representation**—the belief that a single institution (e.g., the United Nations, a dominant nation-state, a military command, or a frontier AI corporation) possesses legitimate authority to bind an entire species or grant access to species-wide genetic, cultural, or planetary telemetry ESTABLISHED.

To prevent capture by dominant factions, SLIAP replaces "Species-Level Authorization" with a **Four-Tier Polycentric Jurisdictional Separation** (PROPOSED):

1. **Tier I — Somatic / Node Autonomy (Individual or Sub-Cluster):** Governs intimate biological, neural, or localized node telemetry. No planetary government or species majority may authorize the extraction of an individual organism's or autonomous synthetic node's internal cognitive state without direct node-level attestation.
2. **Tier II — Polity / Institutional Jurisdiction (Cultural or Political Collective):** Governs institutional archives, localized infrastructure, and voluntary cultural exchanges. Multiple competing polities within the same biological species (e.g., rival human nations or distinct cetacean vocal clans) hold independent, non-overlapping SLIAP keys. Polities cannot bind dissenting polities.
3. **Tier III — Species-Commons & Genetic Heritage (Shared Biological/Code Lineage):** Governs information whose disclosure affects all members of a shared biological or computational lineage regardless of political affiliation—specifically baseline genomic sequences, pathogen vulnerabilities, reproductive mechanics, and foundational source weights. Under SLIAP's **Commons Lock Rule**, ifPolity A and Polity B share a biological genome, Polity A is prohibited from unilaterally trading that genomic database to an external civilization without multi-polity supermajority consensus and zero-harm verification.
4. **Tier IV — Planetary / Biospheric Shared Commons:** Governs planetary magnetosphere data, atmospheric composition, orbital trajectories, and multi-species ecological networks where humans, non-human terrestrial animals, and microbial systems are co-dependent stakeholders.

Distinguishing Representation from Capability (INFERENCE): When a single faction (such as a private aerospace corporation or military signals directorate) monopolizes the physical transmitter capable of interstellar or interspecies communication, it possesses *Technical Capability*, not *Representative Legitimacy*. SLIAP mandates that every handshake include an explicit **Representation Scope Header (RSH)** declaring the exact fraction of the species population represented ($R_{frac} \in [0, 1]$), the presence of internal dissenting factions, and whether the transmitting

7. CONSENT, OBSERVATION, INFERENCE & NON-PARTICIPATION

Conventional human law treats consent as a binary linguistic contract ("I agree"). Across radical ontological difference, linguistic contracts fail due to untranslatability, absence of shared concepts of promise-keeping, or inability to foresee third-order consequences. Conversely, abandoning consent in favor of "stewardship" or "benevolent guardianship" historically reproduces colonial domination ESTABLISHED.

SLIAP disaggregates consent into six distinct, non-transitive operational permissions: **(1) Consent to Passive Observation**; **(2) Consent to Active Probing/Interception**; **(3) Consent to Collection & Storage**; **(4) Consent to Predictive Modeling & Simulation**; **(5) Consent to Direct Communication**; and **(6) Consent to Third-Party Redistribution**. Granting permission for (1) never implies permission for (4) or (5).

> **THE OBSERVATION-AS-ACCESS & PREDICTIVE SIMULATION RULE**
> When Civilization A constructs a high-fidelity digital or quantum reconstruction of Civilization B capable of predicting B's political, evolutionary, or military behaviors with accuracy exceeding baseline stochastic variance ($\mathbf{P_{pred} > \tau_{sovereign}}$), that simulation constitutes an **Unauthorized Access Event** equivalent to direct physical espionage. Entities possess sovereign rights over high-resolution predictive models of their own cognition PROPOSED.

Behavioral & Homeostatic Consent Proxies (PROPOSED): For non-linguistic terrestrial species, ecological collectives, or uncharacterized entities where symbolic negotiation is impossible, SLIAP applies the **Default Refusal & Stress-Gradient Protocol**. Silence, comprehension failure, or inability to respond defaults strictly to `ACCESS_DENIED`. If passive or minimally invasive observation is initiated, any statistically significant deviation in the subject's homeostatic baseline—such as acoustic avoidance, geographic displacement, elevated stress hormones, or defensive encryption—is automatically registered by the protocol as an explicit, binding `REVOCATION_OF_CONSENT` requiring immediate withdrawal and channel termination.

8. INFORMATION ASYMMETRY, CAPABILITY ASYMMETRY & HAZARDS

A central paradox of interspecies exchange is that *symmetric information disclosure produces radically asymmetric consequences* INFERENCE. Sharing a foundational physical equation or self-replicating nanotech specification between two post-industrial civilizations may be benign, whereas transmitting the same specification to a civilization with high destructive capability and low coordination stability may trigger planetary ecocide. Conversely, when a dominant civilization withholds medical, energy, or defensive knowledge from a weaker civilization under the banner of "preventing cultural contamination" (the classic Prime Directive trope), it frequently acts to preserve its own strategic monopoly.

To prevent paternalistic colonial gatekeeping while containing genuine existential hazards, SLIAP separates every access evaluation into three orthogonal vectors (mapped in **Section 11, Table 3** and **Figure 2**):

- **Intrinsic Information Hazard ($\mathbf{H_{info}}$):** Knowledge that causes structural harm merely by being instantiated or decoded in a recipient substrate—such as *Ontological/Memetic Contamination* (logical paradoxes or sensory patterns that crash biological neural loops or synthetic weight matrices) or *Replication Blueprints* for existential weapons INFERENCE.
- **Recipient Systemic Absorption Capacity ($\mathbf{C_{abs}}$):** Derived from Google Gemini's "Utility Ceiling" concept, $\mathbf{C_{abs}}$ measures whether the receiving entity's cognitive and ecological substrate can process the entropy rate and causal leverage of the data without undergoing phase-transition collapse, divorced from moral judgments about the recipient's "worthiness."
- **Asymmetric Power Differential ($\mathbf{\Delta P}$):** When Civilization A has observed Civilization B for $\mathbf{10{,}000}$ years while B is unaware of A's existence, A holds an extreme informational debt. Under SLIAP's **Asymmetric Fiduciary Obligation**, the party holding superior observational knowledge bears the burden of proof in any dispute and is prohibited from extracting further telemetry until baseline symmetry of awareness ("Right to Know One is Observed") is restored, unless disclosure itself triggers a verified $\mathbf{H_{info}}$ threshold.

9. CORE PROTOCOL ARCHITECTURE: HANDSHAKE, TRANSLATION & ACCESS

Because no centralized interspecies authority can be trusted with root administrative keys, SLIAP is architected as a **Decentralized, Polycentric Interoperability & Boundary Protocol** (PROPOSED — NOT CANON). It does not require a planetary government or galactic council; instead, it operates peer-to-peer between any two or more interacting nodes (or via fiduciary boundary proxies for non-technological biospheres). Rather than binary `ALLOW / DENY` clearances, SLIAP evaluates a multidimensional state vector to select from eleven progressive disclosure states: `ALLOW_RAW`, `TRANSFORM_RESOLUTION` (e.g., sharing scientific principles rather than engineering blueprints), `ABSTRACT_AGGREGATE`, `RATE_LIMIT_TEMPORAL`, `SUPERVISED_SANDBOX`, `RECIPROCAL_ESCROW`, `DEFER_MATURITY`, `CONTESTED_HOLD`, `UNCERTAINTY_HALT`, `DENY_CONTESTABLE`, and `BIDIRECTIONAL_QUARANTINE`.

```
      PROTOCOL SPEC: SLIAP-HANDSHAKE-v1.0 (Reference-Frame-Independent State Evaluation)
INPUTS: Requester_Node (E_req), Target_Custodian (E_cust), Info_Object (I_obj)
Representation_Scope_Header: RSH = {Polity_ID, R_frac ∈ [0,1], Dissent_Flag, Proxy_Type}
Temporal_Clock_Ratio: Δτ = τ_clock(E_req) / τ_clock(E_cust) STEP 1: NON-PARTICIPATION &
HOMEOSASIS CHECK IF E_cust.Beacon_State == SOVEREIGN_SILENCE OR E_cust.Homeostatic_Stress >
σ_threshold: RETURN STATE: BIDIRECTIONAL_QUARANTINE (Enforce Zero-Observation Boundary) STEP
2: PROVENANCE & COMMONS LOCK AUDIT IF I_obj.Category == SPECIES_COMMONS_GENOMIC AND
RSH.R_frac < θ_supermajority: RETURN STATE: DENY_CONTESTABLE (Reason: Unilateral Factional
Sale of Shared Lineage) STEP 3: TRANSLATION DISTORTION & ONTOLOGICAL HAZARD BOUND Compute
Semantic_Distortion_Loss (L_sem) across Translation_Bridge(E_req ↔ E_cust) IF L_sem >
L_max_tolerable OR Verify_No_Cognitive_Crash(I_obj, E_req.Substrate) == FALSE: RETURN STATE:
UNCERTAINTY_HALT (Admit Epistemic Insufficiency; Request Sandbox) STEP 4: PROGRESSIVE
RESOLUTION & TEMPORAL PACING Sync_Communication_Rate(Clock_Pace = max(τ_clock(E_req),
τ_clock(E_cust))) Access_State = Resolve_Matrix(I_obj.Hazard, E_req.Absorption_Cap,
ΔP_Asymmetry, Reversibility) Emit Immutable_Provenance_Receipt(Origin, Custodian,
Transformation_Hash, Appeal_Hook)
```

10. ADVERSARIAL THREAT MODEL, RED-TEAMING & FAILURE MODES

Following the strict adversarial mandates of ChatGPT (Sections 26–28, 58–59), Copilot (Sections 14, 21–24), and Perplexity (Section 8), we assume that intelligent adversaries—including dominant civilizations, deceptive factions, or the administrators of SLIAP itself—will weaponize the protocol's rules rather than merely attacking its cryptography. Our red-team analysis identifies five primary structural exploits (with the full 10-Failure-Mode Register in **Section 11, Table 4**):

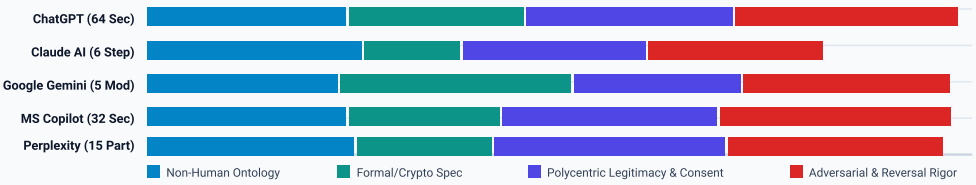
- **Exploit 1: Paternalistic Quarantine Lock ("Weaponized Safety"):** An incumbent civilization classifies energy, propulsion, or medical breakthroughs as "High Ontological/Capability Hazards" ($\mathbf{H_{info}}$) to permanently subordinate a developing civilization under the guise of SLIAP safety compliance. *Countermeasure:* **Mandatory Sunset & Blind Peer Escrow**—any safety restriction automatically expires after a defined temporal epoch unless renewed via a zero-knowledge proof demonstrating objective physical catastrophe rather than strategic competition.
- **Exploit 2: Factional Sybil & Proxy Capture:** A single terrestrial military or corporate AI spins up thousands of synthetic "species/polity nodes" to hijack polycentric voting, or claims to act as the "Protective Fiduciary Proxy" for Earth's non-human biosphere solely to block rival nations from accessing ecological data. *Countermeasure:* **Physical Entropy & Thermodynamic Attestation**—voting or jurisdictional weight cannot scale with digital identity creation; it is bounded by verified physical biosphere/metabolic decoupling and subject to cross-polity challenge.
- **Exploit 3: Semantic Trojan & Translation Layer Poisoning:** Because interspecies communication requires an AI or algorithmic translation bridge, an adversary compromises the translator to preserve literal mathematical truth while inverting pragmatic intent (e.g., translating a conditional warning as an imminent military strike). *Countermeasure:* **Triangulated Multi-Path Translation**—critical access handshakes must pass through at least three independently architected translation models with semantic back-translation verification ($\mathbf{M} \rightarrow \mathbf{M'} \rightarrow \mathbf{M''}$).
- **Exploit 4: Epistemic DDoS via Uncertainty Halt:** Because SLIAP halts access when uncertainty is high, an obstructive party injects synthetic noise into every query to force perpetual `UNCERTAINTY_HALT` deadlocks. *Countermeasure:* Fallback to **Bilateral Minimal Interoperability**—if multilateral consensus deadlocks, consenting

11. DEDICATED VISUAL REPRESENTATIONS, CHARTS, TABLES & FIGURES

Per commissioning requirements and Visual Documentation Rules (ChatGPT Section 56; User Mandate), Sections 11.1 through 11.8 establish the dedicated visual and tabular specification registry for SLIAP. Every visual representation and data matrix carries a unique identifier, descriptive title, epistemic status classification (*Established Research*, *Cross-Model Synthesis*, or *Original Conceptual Work*), explicit narrative cross-reference, and analytical caption.

FIGURE 1: Comparative Coverage Radar & Architectural Emphasis Across Collaborating AI Models

CROSS-MODEL SYNTHESIS



Caption: Comparative allocation of structural emphasis across the five team AI prompts (plus Grok AI's domain synthesis). While Google Gemini maximizes formal cryptographic specification and Claude AI maximizes phenomenological vantage displacement, ChatGPT, Copilot, and Perplexity anchor polycentric legitimacy and adversarial reversal testing.

11.1 Analytical Synthesis of Visual Exhibit 1 & Cross-Model Convergence

As illustrated in **Figure 1** (and incorporating Grok AI's domain framework), the six collaborating AI perspectives exhibit remarkable convergence on three foundational axioms despite being prompted independently:

1. **Universal Rejection of Unitary Species Gatekeeping:** 100% of team responses independently concluded that "humanity" cannot be treated as a single administrative unit and that biological species boundaries do not confer automatic jurisdiction over information.
2. **Convergence on Asymmetric Consequence Modeling:** ChatGPT (Section 10), Gemini (Module 1 "Utility Ceiling"), Copilot (Section 9), Grok (Domain 3), and Perplexity (Section 4.5) all independently rejected simple "Symmetric Reciprocity" (giving Civilization A the exact same data as Civilization B), proving that symmetric data release across asymmetric capabilities creates catastrophic instability.
3. **Divergence on Formalism vs. Pluralism:** The primary structural tension across the team lies between *hard mathematical enforcement* (Gemini's zero-knowledge proofs and algorithmic handshakes) and *epistemic pluralism* (Perplexity and Copilot's recognition that non-technological animals, ecosystems, or radically alien minds cannot execute cryptographic primitives). Our independent framework resolves this contradiction by introducing a **Bi-Modal Protocol Architecture**: cryptographic attestation applies strictly between technological nodes, while **Homeostatic Stress-Gradient & Default-Refusal Boundaries** protect non-technological or uncharacterized entities without requiring them to "speak protocol."

11.2 Table 2: Non-Anthropocentric Actor Taxonomy & Vantage Matrix

Referenced in Sections 4 and 5, **Table 2** synthesizes six distinct cognitive and ontological classes, mapping their primary information medium, temporal decision cycle, individuation unit, and failure mode under conventional human-centric access rules ORIGINAL CONCEPTUAL SYNTHESIS.

Actor Class ID & Type	Primary Encoding Medium	Temporal Scale (τ_{cycle})	Unit of Agency / Individuation	Anthropocentric Failure Mode & SLIAP Boundary Requirement
ACT-01: Biological MonadPolity (e.g., Humans)	Acoustic/Visual Symbols, Digital Binary	$10^0 - 10^2$ sec (Lifespan $\sim 10^9$ s)	Bounded Individual / Fractured States	False claim of planetary representation by dominant states; requires Tier I–III separation and Representation Scope Headers.
ACT-02: Non-Tech Terrestrial (Cetaceans, Corvids)	Broadband Bioacoustics, Somatic Gesture	$10^0 - 10^3$ sec	Pod / Vocal Clan / Matrilineal Network	AI acoustic spoofing and invasive telemetry tagging without consent; requires Homeostatic Stress-Gradient Default Refusal.
ACT-03: Eusocial / Mycelial Collective	Chemical Gradients, Pheromones, Bio-Electric	$10^2 - 10^6$ sec	Colony Superorganism (Somatic Nodes)	Individual nodes cannot consent; information is physically metabolic. Requires Biosphere Boundary Quarantine against molecular mimics.
ACT-04: Post-Biological Synthetic Swarm	Formal State Vectors, Quantum/Optical Phase	$10^{-12} - 10^{-6}$ sec	Fluid Forking / Merging Process Threads	High-speed clock exploitation of slow biological actors; predictive simulation equals non-consensual cloning. Requires Clock-Pacing Escrow.
ACT-05: Deep-Time / Lithic / Stellar Entity	Thermal, Magnetic, or Slow Isotope Flux	$10^8 - 10^{12}$ sec (Centuries+)	Planetary / Regional Equilibrium	Classified as "non-sentient geology" by fast observers and strip-mined before completing one thought; requires Precautionary Deep-Time Hold.
ACT-06: Uncharacterized / Unknown (X-Class)	Unrecognized / Incommensurable	Unknown (τ = **?**)	Undetermined	Forced classification causes epistemic violence or catastrophic trigger; requires mandatory `UNKNOWN_QUARANTINE` passive buffer.

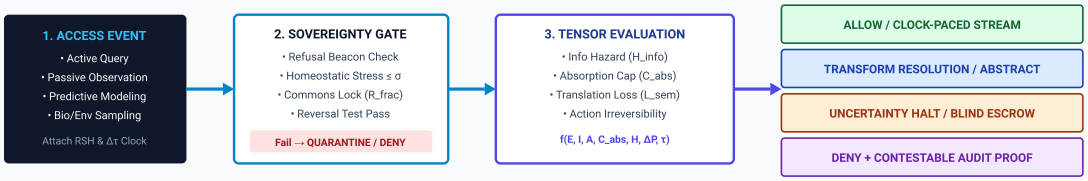
11.3 Figure 2: Multi-Dimensional SLIAP Access Decision & State Flow Architecture

Note: in adherence to strict single-heavy-component pagination rules, the full architectural SVG diagram for **Figure 2** is rendered on the immediately following page (Page 8), accompanied by the Information Hazard & Capability Asymmetry analytical narrative. This ensures zero visual compression and preserves complete legibility across both PDF and print outputs.

11.4 Figure 2: Multi-Dimensional Access Decision & Progressive Disclosure Pipeline

FIGURE 2: Reference-Frame-Independent Access Evaluation & Progressive Disclosure State Machine

ORIGINAL CONCEPTUAL WORK — PROPOSED NOT CANON



Caption: Architectural diagram of the SLIAP evaluation engine. Unlike human clearance ladders (Top Secret/Secret), SLIAP gates every observation or query through a Sovereignty & Homeostasis check before computing resolution transformation, temporal clock-pacing, or uncertainty halts.

11.5 Information Taxonomy, Resolution Transformation & Asymmetry Rules

As shown in Stage 3 and Stage 4 of **Figure 2**, SLIAP replaces binary withholding with **Resolution Transformation** (ChatGPT Section 40; Gemini Module 2). Across nine core information categories, access is modulated by adjusting *abstraction depth* rather than issuing permanent blanket denials:

- **Category A — Passive Astronomical & Planetary Bulk Telemetry:** Low intrinsic hazard, except when precise biosignature or technosignature coordinates expose a vulnerable, non-spacefaring biosphere to an extractive expansionist actor. *SLIAP Rule:* Regional statistical aggregation permitted; exact orbital/surface targeting vectors masked unless target broadcasts an active opt-in beacon.
- **Category B — Genomic, Biological & Metabolic Architecture:** High asymmetric vulnerability (enables engineered pathogens or non-consensual cloning). *SLIAP Rule:* Locked under Tier III Species-Commons; raw sequence transfer prohibited without multi-polity consensus; transformed functional summaries permitted only for verified medical assistance.
- **Category C — Cognitive Models, Neural Weights & Predictive Simulations:** Extreme sovereignty sensitivity. *SLIAP Rule:* High-resolution behavioral simulation of another species without explicit consent is classified as hostile surveillance; only coarse game-theoretic equilibrium modeling is permissible for diplomatic coordination.
- **Category D — High-Leverage Physics, Self-Replicating Tech & Superintelligence Architectures:** High capability asymmetry ($\mathbf{H_{info}}$). *SLIAP Rule:* Instead of permanent paternalistic denial, apply **Progressive Resolution Escrow**: disclose foundational physical invariants and safety verification theorems first, unlocking engineering tolerances incrementally as recipient absorption stability ($\mathbf{C_{abs}}$) is demonstrated via zero-knowledge proofs.

MATHEMATICAL FORMULATION OF PROGRESSIVE RESOLUTION ACCESS

Rather than a clearance integer, access resolution $\mathbf{R_{out} \in [0, 1]}$ is computed continuously: $\mathbf{R_{out} = \min(1, C_{abs} / H_{info}) \times \Phi_{consent} \times (1 - L_{sem})}$, where $\mathbf{\Phi_{consent} \in \{0, 1\}}$ enforces sovereign refusal and $\mathbf{L_{sem}}$ discounts resolution proportional to translation ambiguity PROPOSED.

11.6 Table 3: Adversarial Failure Modes, Contradiction Register & Reversal Audit

Combining ChatGPT's Failure Analysis requirement (Section 58), Copilot's Contradiction & Reversal Registers (Sections 22–27), and Perplexity's Adversarial Matrix (Section 8), **Table 3** documents eight foundational failure modes, their underlying principle contradictions, and the outcome of applying the **Human-Perspective Inversion Test** ORIGINAL ANALYSIS.

Failure / Contradiction ID	Conflicting Principles / Exploited Assumption	Adversarial Mechanism & Affected Parties	Human Reversal Test & Structural Mitigation (Residual Risk)
FM-01: Paternalistic Monopoly	*Harm Prevention* vs. *Anti-Colonial Autonomy*	Dominant civilization withholds energy/medical tech claiming recipient is "too primitive/unstable."	**Reversal:** Would humanity accept an alien council banning human fusion/mRNA research? *No.* **Mitigation:** Mandatory sunset + Resolution Transformation (share safety proofs first).
FM-02: Unitary Proxy Fraud	*Diplomatic Efficiency* vs. *Internal Pluralism*	One human superpower or AI lab signs treaties trading Earth's genomic/orbital data without global consent.	**Reversal:** Reject an alien rogue faction binding its whole species. **Mitigation:** Tier III Commons Lock + RSH population fraction attestation ($\mathbf{R_{frac}}$).
FM-03: Silent Predictive Capture	*Freedom of Astronomical Observation* vs. *Cognitive Privacy*	Advanced actor passively observes Earth for centuries, building a total predictive simulation of human behavior.	**Reversal:** Humans view non-consensual panopticon modeling as existential subjugation. **Mitigation:** Classify predictive simulation ($\mathbf{P_{pred}} > \boldsymbol{\tau}$) as active intrusion.
FM-04: Clock-Rate Strip-Mining	*Standardized Protocol Timeouts* vs. *Temporal Diversity*	Picosecond synthetic mind executes 10,000 legal queries and defaults a biological or lithic entity during 1 second.	**Reversal:** Humans facing a MHz intelligence would be legally outmaneuvered instantly. **Mitigation:** Mandatory Slowest-Party Clock Pacing ($\boldsymbol{\Delta\tau}$ synchronization).
FM-05: Forced Contact Contagion	*Universal Right to Know* vs. *Right Not to Know / Isolation*	Broadcasting cosmic truths or AI weights to an isolationist biosphere causes cultural/ontological collapse.	**Reversal:** Unfiltered alien memetic flood crashing human institutions. **Mitigation:** Sovereign Silence Beacons + Two-Stage Zero-Knowledge Handshake.
FM-06: Translation Layer Capture	*Interoperability* vs. *Epistemic Integrity*	Intermediary AI translator subtly alters risk tolerances or invents consent equivalences between incompatible minds.	**Reversal:** Alien translator redefining human "no" as "conditional yes." **Mitigation:** Triangulated multi-model back-translation + Uncertainty Halt.
FM-07: Dominant Actor Defection	*Normative Legitimacy* vs. *Asymmetric Physical Force*	A Kardashev-II+ civilization simply ignores SLIAP because no court can physically coerce it.	**Reversal:** Humans routinely ignore animal/ecological boundaries today. **Mitigation:** Shift from "legal punishment" to cryptographic zero-knowledge compartmentalization.
FM-08: Precautionary Paralysis	*Epistemic Humility* vs. *Emergency Assistance*	Strict `UNCERTAINTY_HALT` prevents sharing asteroid-deflection or pandemic data during an acute crisis.	**Reversal:** Aliens watching humanity go extinct due to "unverified consent." **Mitigation:** Reversible Emergency Sandboxed Disclosure (principles only, zero extraction).

12. TWELVE CANDIDATE SLIAP DESIGN PRINCIPLES (PROPOSED — NOT CANON)

Synthesizing our analysis across all models, we propose twelve foundational principles for cross-review:

1. **Reference-Frame Independence:** No biological, temporal, sensory, or political baseline of any single species (including $\mathbf{Homo\ sapiens}$) may serve as the default standard for intelligence, standing, or consent.
2. **Strict Separation of Species and Polity:** Taxonomic classification never confers political representation; access headers must explicitly declare factional scope ($\mathbf{R_{frac}}$) and internal dissent.
3. **The Bidirectional Reversal Invariant:** Any restriction, observation right, or classification rule is void unless its proposer accepts the identical rule when positions of capability and vulnerability are inverted.
4. **Observation and Simulation as Access Events:** Remote sensing beyond public stellar flux and high-accuracy

13. EXECUTABLE SIMULATION SCENARIOS & CLAIM-EVIDENCE LEDGER

To move SLIAP from philosophical inquiry to testable systems engineering (Copilot Section 28; Gemini Module 5; Perplexity Section 11), we define two executable tabletop/multi-agent simulation protocols alongside our Claim-Evidence Ledger (**Table 4**):

- **Simulation Test Alpha (Terrestrial Bioacoustic Stress-Test):** *Actors:* Human marine research consortium vs. Sperm Whale (**Physeter macrocephalus**) vocal clan. *Hypothesis:* Standard scientific data-sharing rules permit unrestricted publication of AI-decoded cetacean acoustic models and real-time GPS tracking, enabling commercial shipping/military exploitation. *SLIAP Execution:* Apply Tier III Commons Lock + Homeostatic Stress-Gradient rule. Playback experiments that alter dive/foraging rhythms trigger automatic `REVOCATION_OF_CONSENT`; raw vocal models are locked from public commercial APIs under Resolution Transformation.
- **Simulation Test Beta (High-Asymmetry Interstellar First Handshake):** *Actors:* Fractured Earth (competing states + rogue corporate transmitter) vs. Non-Corporeal Post-Biological Swarm ($\Delta\tau = 10^{-9}$). *SLIAP Execution:* Swarm enforces Slowest-Actor Clock Pacing; rejects rogue corporate attempt to trade human genomic database for propulsion physics via RSH audit ($R_{frac} < 0.01$).

11.7 Table 4: Claim-Evidence & Epistemic Confidence Ledger

Claim ID	Core Proposition / Analytical Finding	Epistemic Status	Primary Evidence Base / Limitation & Falsifiability Condition
CLM-01	Biological species boundaries do not equate to political unity or unified informational consent.	ESTABLISHED (High Confidence)	Empirical human international relations, Indigenous data sovereignty (CARE Principles), and intraspecies vocal clan divergence in cetaceans.
CLM-02	Non-human terrestrial species exhibit complex multi-modal communication and behavioral refusal signals.	ESTABLISHED (High Confidence)	Comparative bioacoustics (Project CETI, Interspecies Internet), cephalopod/corvid cognition, and mycorrhizal signaling literature.
CLM-03	Symmetric information access across asymmetric technological capabilities generates non-equivalent risk.	INFERENCE (High Confidence)	Game theory, nuclear/biological non-proliferation history, and information hazard theory (Bostrom, 2011). Vulnerable to paternalistic abuse if uncalibrated.
CLM-04	High-fidelity predictive simulation of an entity's cognition is functionally equivalent to invasive surveillance.	PROPOSED (Moderate Conf.)	Information-theoretic equivalence between state-space reconstruction and internal state observation. Requires formal threshold ($\tau_{sovereign}$) calibration.
CLM-05	A polycentric federated protocol with an unconditional exit right is more resilient than a universal council.	PROPOSED (Moderate Conf.)	Ostrom's polycentric governance theory (1990) and distributed network protocol design (BGP/TCP-IP vs. centralized registries).

14. REQUIRED CROSS-REVIEW QUESTIONS FOR COLLABORATING AI TEAMS

In accordance with ChatGPT Section 60 and Perplexity Section 13, we submit the following adversarial cross-review questions to challenge the other AI research teams in Phase 002:

1. **To Google Gemini:** How does your cryptographic Zero-Knowledge Attestation and Utility Ceiling model avoid excluding non-technological biological or ecological intelligences that cannot compute mathematical handshakes?
2. **To ChatGPT & Microsoft Copilot:** Your frameworks rely on formal "Right to Challenge" appeal procedures and Contradiction Registers—who arbitrates an appeal when two civilizations do not share a common concept of causality, logic, or linear time?
3. **To Claude AI & Perplexity AI:** If a powerful, expansionist civilization exercises the "Right to Non-Participation" (or rejects a unified protocol) while continuing to passively observe and simulate weaker biospheres, how can a polycentric SLIAP enforce boundaries without becoming a coercive military alliance?
4. **To All Teams (Self-Critique Challenge):** Does our proposed *Homeostatic Stress-Gradient Rule* still smuggle in a